DATA SCIENCE • MODULE 07

Hypothesis Testing

Make Data-Driven Decisions With Statistical Confidence

■ STATISTICAL TESTING

■ LEARNING OUTCOME

By the end of this PDF you will design, execute, and correctly interpret hypothesis tests to validate business decisions — from A/B tests to clinical trials to product analytics.

■ Skills You Will Gain
• Formulate null and alternative hypotheses correctly
• Choose the right statistical test for any situation
• Run t-tests, chi-square, ANOVA, and Mann-Whitney
• Understand Type I & II errors, power, and sample size
• Design and analyze A/B tests from scratch
• Report results in business language

■ **Hypothesis testing is how Netflix decides which thumbnail to show, how Amazon chooses button colors, and how pharma companies approve drugs. This skill is everywhere.**

SECTION 1

The Hypothesis Testing Framework

1. **Define Hypotheses**
   H■ (null): No effect/difference exists H■ (alternative): An effect/difference exists

2. **Choose α (significance level)**
   α = 0.05 (5% false positive rate) is industry standard Use α=0.01 for medical/financial decisions

3. **Choose the right test**
   Based on: data type, number of groups, distribution shape

4. **Calculate test statistic**
   t-score, z-score, chi-square, F-statistic depending on test

5. **Get p-value**
   Probability of observing this result if H■ is true

6. **Make decision**
   p < α → Reject H■ (result is significant) p ≥ α → Fail to reject H■ (insufficient evidence)

7. **Report finding**
   Business language: 'Version B increased conversion by 12% (p=0.02)'

SECTION 2

Choosing the Right Test

Scenario	Test	Function
Compare mean to known value	One-sample t-test	stats.ttest_1samp(data, popmean)
Compare means: 2 independent groups	Two-sample t-test	stats.ttest_ind(a, b)
Compare means: 2 paired groups	Paired t-test	stats.ttest_rel(before, after)
Compare means: 3+ groups	One-way ANOVA	stats.f_oneway(g1,g2,g3)
Non-normal, 2 groups	Mann-Whitney U	stats.mannwhitneyu(a,b)
Non-normal, 3+ groups	Kruskal-Wallis	stats.kruskal(g1,g2,g3)
Two categorical variables	Chi-square test	stats.chi2_contingency(table)
Proportion comparison	Z-test for proportions	proportions_ztest(count,nobs)
Normality check	Shapiro-Wilk / K-S	stats.shapiro(data)

SECTION 3

Running Tests in Python

```
from scipy import stats import numpy as np # ■■ Two-sample t-test ■■ group_A =
df[df['Group']=='A']['Conversion'] group_B = df[df['Group']=='B']['Conversion'] t, p =
stats.ttest_ind(group_A, group_B, equal_var=False) # Welch's t-test print(f't={t:.3f}, p={p:.4f},
significant={p<0.05}') # ■■ Chi-square test ■■ ct = pd.crosstab(df['Gender'], df['Purchased'])
chi2, p, dof, expected = stats.chi2_contingency(ct) print(f'chi2={chi2:.3f}, p={p:.4f}') # ■■
One-way ANOVA ■■ groups = [df[df['Region']==r]['Sales'] for r in df['Region'].unique()] f, p =
stats.f_oneway(*groups) print(f'F={f:.3f}, p={p:.4f}') # ■■ Check normality first ■■ stat, p_norm
= stats.shapiro(group_A) if p_norm < 0.05: print('Non-normal → use Mann-Whitney') u, p_mw =
stats.mannwhitneyu(group_A, group_B, alternative='two-sided')
```

SECTION 4

Type I & II Errors + Statistical Power

	H■ True	H■ False
Reject H■	Type I Error (α) — False Positive 'We said it works but it doesn't'	Correct Decision ■ (Power = 1-β)
Fail to Reject H■	Correct Decision ■	Type II Error (β) — False Negative 'We missed a real effect'

```
# Calculate required sample size for desired power from statsmodels.stats.power import
TTestIndPower analysis = TTestIndPower() n = analysis.solve_power(effect_size=0.5, # Cohen's d
power=0.80, # 80% chance of detecting real effect alpha=0.05) # 5% false positive rate
print(f'Need {int(np.ceil(n))} samples per group')
```

SECTION 5

A/B Testing End-to-End

1 **Define Metric**
Primary: conversion rate. Secondary: revenue per user, session time

2 **Calculate Sample Size**
Use power analysis: effect_size=0.05 (5% lift), power=0.8, α=0.05

3 **Run Experiment**
Randomly split users 50/50, run for full business cycle (≥1 week)

4 **Check for Bias**
Is randomization working? Compare baseline demographics of A vs B

5 **Run t-test or z-test**
proportions_ztest([conv_A,conv_B],[n_A,n_B])

6 **Check practical significance**
Statistical significance ≠ business significance. Is 0.1% lift worth the cost?

7	**Report** 'Version B increased conversion from 3.2% to 3.8% (+18.75%), p=0.012 — implement'

```
# A/B test with proportions from statsmodels.stats.proportion import proportions_ztest conversions
= [120, 145] # B converted more nobs = [2000, 2000] # same sample size z, p =
proportions_ztest(conversions, nobs) print(f'z={z:.3f}, p={p:.4f}') lift = (145/2000 - 120/2000) /
(120/2000) * 100 print(f'Lift: {lift:.1f}%')
```

SECTION 6

Effect Size — Why p-value Alone Is Not Enough

Effect Size Measure	Formula	Interpretation
Cohen's d (t-test)	d = (μ■-μ■) / pooled_std	0.2=small, 0.5=medium, 0.8=large
Cramer's V (chi-square)	$V = \sqrt{\chi^2/n \cdot \min(r-1,c-1)}$	0.1=small, 0.3=medium, 0.5=large
η^2 (ANOVA)	η^2 = SS_between / SS_total	Proportion of variance explained

■ PRACTICE QUESTIONS

Q1. What does p-value = 0.03 mean?

> ***Answer:*** *If the null hypothesis were true, there's only a 3% chance of observing a result this extreme by chance. Since 0.03 < 0.05, we reject H■.*

Q2. You want to compare revenue across 4 sales regions. Which test do you use?

> ***Answer:*** *One-way ANOVA — it compares means across 3 or more independent groups simultaneously.*

Q3. What is Type I error and how do you control it?

> ***Answer:*** *False positive — rejecting H■ when it's actually true. Controlled by the significance level α. Setting α=0.05 means you accept a 5% false positive rate.*

Q4. Why is sample size calculation important before running an A/B test?

> ***Answer:*** *Too small = underpowered test (miss real effects). Too large = waste of time and money. Power analysis finds the minimum sample needed to detect a meaningful effect.*

Q5. Statistical test shows p=0.04 for a 0.01% conversion lift. Should you ship the change?

> ***Answer:*** *No — it's statistically significant but practically insignificant. Always check effect size and business impact alongside p-value.*

■ Hypothesis Testing Cheat Sheet	
`H■: no effect`	Null hypothesis — default assumption
`H■: effect exists`	Alternative hypothesis — what you test
`α = 0.05`	5% significance level (standard)
`p < α → reject H■`	Result is statistically significant
`stats.ttest_ind(a,b)`	2-group mean comparison
`stats.ttest_rel(b,a)`	Paired before/after test
`stats.f_oneway(*groups)`	ANOVA: 3+ groups
`stats.chi2_contingency(t)`	Chi-square: categorical
`stats.mannwhitneyu(a,b)`	Non-parametric 2-group

`stats.shapiro(data)`	Normality check
`proportions_ztest(c,n)`	A/B test proportions
`Cohen's d = 0.8`	Large effect size